temperature. Above this temperature enzymes get destroyed and below this they remain inactive (Figure 9.3a). The second important factor is pH. Enzymes work at a specific pH known as optimum pH. For example, digestive enzymes work well at acidic pH and intracellular enzymes work better at neutral pH (Figure 9.3b). The other factors are enzyme and substrate concentration, if the initial concentration is high it gradually increases the reaction rate (Figure 9.3c).

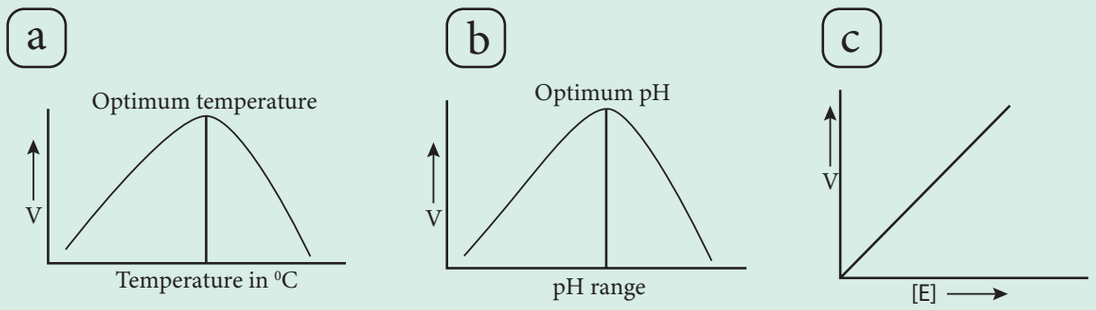


Figure 9.3 Factors affecting enzyme activity

9.1 DERIVATION OF MICHAELIS - MENTEN EQUATION

To understand the mechanism of any catalytic reaction, one should study the kinetic behaviour of the reaction systems, where the rate of the reaction can be obtained at various concentrations of the enzyme and the substrate. Formation of enzyme substrate complex is a critical event in an enzyme-catalyzed reaction. Michaelis and Menten proposed that the binding of the substrate and the enzyme is reversible and derived a kinetic model for a simple single substrate enzymatic reaction. This laid the basis for understanding the various factors that influence the rate of the reaction, such as concentration of substrate, temperature and pH of the system.

The kinetics of enzyme catalyzed reactions is explained by Michaelis- Menten equation. It can be derived with the following assumptions:

1. The concentration of enzyme is much lower than the concentration of the substrate.
2. The initial rate (or initial velocity), designated v_0, when [S] is much greater than the concentration of enzyme [E], is measured
3. None of the products formed revert to the initial substrate.
4. A steady-state condition is reached between the formation of ES complex and its decomposition.

The enzyme (E) combines with the substrate (S) to form ES complex with the rate constant k_1. The ES complex can dissociate either into 'E' and 'S' with the rate constant k_2 or it can form the product (P) with the rate constant k_3.

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